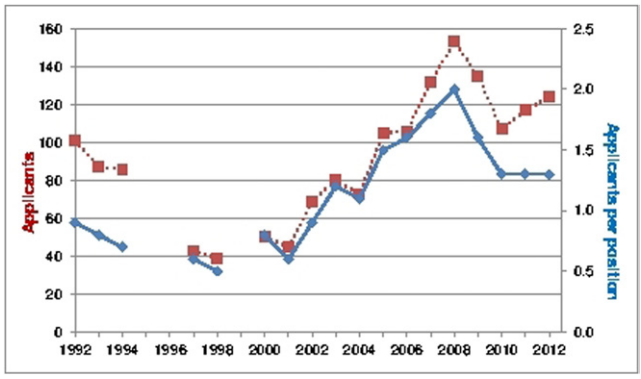
in 1992. Numbers of all fellowship programs and positions and all applicants and positions filled were recorded annually. We plotted applicant and position data by year to examine trends over time.

RESULTS: The number of applicants has tripled since 1996 with more applicants and applicants per position than ever before. The numbers of participating programs, positions offered, and applicants declined during the transition period between 1997 and 2003. The number of participating fellowship programs rose steadily thereafter and is now equivalent to numbers before instituting the additional training year. While the number of positions increased in recent years, it has not reached levels before 1996. In the last nine years, the competition for acceptance into a fellowship in MFM has risen; there has been a mean of 42.2 ± 17.1 unmatched applicants per year. Comparatively, from 1992-1996 and during the expansion period (1997-2003) the mean number of unmatched applicants was 14.7 ± 4.5 and 12.3 ± 5.9 respectively. Similarly, from 2004-2012 there has been a mean of 3.8 ± 3.0 unfilled programs per year versus 28.7 ± 2.6 from 1992-1996 and 23.7 ± 10.7 from 1997-2003.

CONCLUSION: The transition from a two to a three year training program in 1996 only temporarily halted the otherwise steady and sustained growth in MFM fellowship programs, positions, and applicants.



540 Training needs in operative obstetrics among first year maternal-fetal medicine fellows

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OBJECTIVE: To estimate the frequency with which 13 operative obstetric procedures were performed by first year MFM fellows during their preceding residencies, and to determine factors associated with increased comfort doing and teaching these procedures during fellowship.

STUDY DESIGN: We used Qualtrics software to administer a web-based survey to all 100 first-year fellows, and exact logistic regression to estimate the odds of comfort in doing and teaching these procedures. We present descriptive data on all 13 procedures and additional detail on three common ones: cesarean hysterectomy, low forceps delivery, and breech extraction of a second twin.

RESULTS: Response rate was 86%; frequency data for all 13 procedures is shown in Table 1. Fellows who completed residency in the Northeast/Midatlantic (N=26) were less likely to report comfort doing or teaching low forceps deliveries (OR 0.21 [0.05,0.78], and 0.20 [0.04,0.85], respectively), while those completing fellowship in the West (N=13) reported more comfort performing breech extraction of a second twin (OR 6.84 [1.24, 51.50]); fellows completing residency in the Southeast formed the referent group (N=26). Fellows reporting completion of the three selected procedures > 5 times each during residency were significantly more likely to report comfort doing and teaching them as fellows. Type of residency program (community/academic) was not significantly associated with reported comfort, although size of program was: fellows from residency programs with 5-9 residents/ year (N=57) were less likely to report comfort teaching breech extraction of a second twin (OR 0.18 [0.03, 0.76]) or low forceps delivery (0.27 [0.07, 1.02]) than were those from programs of less than 5 residents per year.

CONCLUSION: The wide range of operative obstetric experience fellows reported gaining in residency varies by program demographic. To attain goals outlined in ABOG's Guide to Learning related to operative obstetrics, many MFM fellows need additional procedural experience.

Operative obstetric frequency data for first year Maternal-Fetal Medicine fellows

Procedure	Times performed during residency(scale: 0 to >25); Median (IQR)	Not comfortable doing or teaching; N (%)	Comfortable doing but not teaching; N (%)	Comfortable doing and teaching; N (%)
Cesarean hysterectomy	3 (1-4)	20 (24)	36 (42)	29 (34)
Breech extraction of a second twin	3 (1-5)	33 (39)	11 (13)	41 (48)
Cerclage: McDonald	10 (5-14)	1 (1)	14 (17)	70 (82)
Cerclage: Shirodkar	0 (0-2)	56 (67)	17 (20)	11 (13)
Cerclage: cervico-isthmic (vaginal approach)	0 (0-0)	80 (94)	4 (5)	1 (1)
Cerclage: abdominal	0 (0-0)	75 (88)	7 (8)	3 (4)
Cerclage: robotic	0 (0-0)	81 (95)	3 (4)	1 (1)
Forceps: outlet	10 (3-15)	28 (33)	11 (13)	46 (54)
Forceps: low	2 (0-7)	43 (51)	11 (13)	31 (36)
Forceps: rotational	0 (0-0)	80 (94)	3 (4)	2 (2)
Forceps: Pipers	0 (0-1)	63 (74)	12 (14)	10 (12)
Vacuum delivery	22 (13- >25)	1 (1)	1 (1)	83 (98)
4th degree laceration repair	3 (2-5)	11 (13)	18 (21)	56 (66)

Column 1 includes data from 86 respondents; columns 2-4 include data from 85 respondents.

541 Estimating blood loss: can an easily accessible visual aid significantly improve visual estimation?

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OBJECTIVE: Obstetric hemorrhage is an emergency situation that requires expedient identification and intervention to prevent maternal and fetal morbidity. Research has shown that visual estimation of blood loss (EBL) is inaccurate and often underestimated, which may lead to delayed treatment. Our objective was to determine whether an easily accessible visual aid would improve provider's performance in the estimation of blood loss.

STUDY DESIGN: We created a pocket card, depicting known volumes of blood on common obstetric materials, to serve as a visual aid in estimating blood loss (Figure). Accuracy of blood loss estimation was assessed at 6 stations with known volumes of artificial blood using materials common to standard delivery kits and recreating common obstetric scenarios. Medical personnel consisting of 151 subjects [obstetrics attending (n=24), residents (n=22), nurses (n=74), midwives (n=11), anesthesia providers (n=13), medical students (n=7)] recorded EBL across a variety of volumes and materials before and after receiving our visual aid. Clinical experience of the evaluators ranged from 0-40 years (Mean 11.54 ± SD 11.89). We prospectively categorized percent error of EBL as underestimation (% error less than −20%), accurate assessment (% error between −20% and +20%); and overestimation (% error more than +20%). Wilcoxon signed rank test was used for analysis.

RESULTS: For all providers, there was a significant improvement in the accuracy of EBL after intervention across all blood volumes tested except for the Chux A (Table). In other analyses, initial accuracy of EBL was significantly affected by provider type in only 2 of the 6 stations (Peri-pad A and Delivery Drape) ($p < 0.05$). This difference